MEDICAL DATA SCIENCE & MACHINE LEARNING

Breast Cancer SVM Classification

Exploratory Data Analysis, Feature Engineering, and Support Vector Machine Diagnostic Pipeline

KEY OVERVIEW

Dataset At A Glance

569

Total Observations

30

Continuous Features

0%

Missing Values

TARGET VARIABLE ANALYSIS

Diagnosis Class Balance

Benign (B) **63.3%**

360 Samples in Dataset

Malignant (M) **36.7%**

209 Samples in Dataset

ARCHITECTURE

End-to-End Classification Flow

STEP 01

Data Ingestion

CSV lookup via Glob.

STEP 02

Preprocessing

StandardScaler zero-mean fit.

STEP 03

SVM Training

RBF Kernel boundary fitting.

STEP 04

Diagnosis

Binary classification result.

METHODOLOGY

Support Vector Machine Optimization

RBF Kernel Selection

Handles complex, non-linear decision boundaries between tumor metric distributions.

Standard Scaling

Prevents features with larger variance from dominating hyperplane margin distance.

Questions & Discussion

Breast Cancer SVM Classification Pipeline Completed Successfully